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#include<stdio.h>
#include<limits.h>
void insert(int ary[], int hFn, int size) {
    int element, pos, n = 0;
    printf("Enter key element to insert\n");
    scanf("%d", &element);
    pos = element % hFn;
    while (ary[pos] != INT_MIN) {
        // INT_MIN and INT_MAX indicate that cell is empty or deleted
        if (ary[pos] == INT_MAX)
            break;
        pos = (pos + 1) % hFn;
        n++;
    }
    if (n == size) {
        printf("Hash table is full. No place to insert this element.\n");
        return;
    }
    ary[pos] = element; // Inserting element
}

void delete(int ary[], int hFn, int size) {
    int element, pos, n = 0;
    printf("Enter element to delete\n");
    scanf("%d", &element);
    pos = element % hFn;
    while (n++ != size) {
        if (ary[pos] == INT_MIN) {
            printf("Element not found in hash table\n");
            return;
        } else if (ary[pos] == element) {
            ary[pos] = INT_MAX; // Mark as deleted
            printf("Element deleted\n");
            return;
        } else {
            pos = (pos + 1) % hFn;
        }
    }
    printf("Element not found in hash table\n");
}

void search(int ary[], int hFn, int size) {
    int element, pos, n = 0;
    printf("Enter element you want to search\n");
    scanf("%d", &element);
    pos = element % hFn;

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while (n++ != size) {
if (ary[pos] == element) {
printf("Element found at index %d\n", pos);
return;
} else if (ary[pos] == INT_MAX || ary[pos] != INT_MIN) {
pos = (pos + 1) % hFn;
}
}
printf("Element not found in hash table\n");
}

void display(int ary[], int size) {
printf("Index\tValue\n");
for (int i = 0; i < size; i++) {
printf("%d\t", i);
if (ary[i] == INT_MIN) {
printf("Empty\n");
} else if (ary[i] == INT_MAX) {
printf("Deleted\n");
} else {
printf("%d\n", ary[i]);
}
}
}

int main() {
int size, hFn, choice;
printf("Enter size of hash table\n");
scanf("%d", &size);
int ary[size];
printf("Enter hash function (if mod 10, enter 10)\n");
scanf("%d", &hFn);
for (int i = 0; i < size; i++)
ary[i] = INT_MIN; // Assigning INT_MIN indicates that cell is empty
do {
printf("\nEnter your choice\n");
printf("1 -> Insert\n2 -> Delete\n3 -> Display\n4 -> Search\n0 -> Exit\n");
scanf("%d", &choice);
switch (choice) {
case 1:
insert(ary, hFn, size);
break;
case 2:
delete(ary, hFn, size);
break;
case 3:

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display(ary, size);  
break;  
case 4:  
search(ary, hFn, size);  
break;  
case 0:  
printf("Exiting...\n");  
break;  
default:  
printf("Enter correct choice\n");  
break;  
}  
} while (choice != 0);  
return 0;  
}
```